

University of Rajshahi
Department of Computer Science and Engineering
B. Sc. Engg. Part2 Odd Semester, Examination-2023 (Affiliated Colleges)
Course: CSE-2121 (Data Structure)
(Session: 2021-22 and 2020-21)

Full Marks: 60

Duration: 3Hours

[Answer six questions taking any three from each section]

SECTION A

1. (a) What is linear array? How can you find number of elements in any linear array? 2.5
(b) Simulate the binary search algorithm on the following data: 11, 22, 33, 44, 55, 66, 77, 88, 99, 110, 121, 132, 143 (Suppose we search for item 120) 4.5
(c) What is 2D array? How can you represent 2D array in memory? 3
2. (a) Describe a situation where storing items in an array is clearly better than storing items on a linked list. 2
(b) One of the advantages of linked list is the ability to insert data into the list easily. Explain with your own words and figures how to insert data at the beginning, after a given node, at the end and to a sorted list. 5
(c) What is overflow and underflow? How can you handle them? 3
3. (a) Suppose that we want to encode a message constructed from the symbols A, B, C, D, E, F and G using a fixed-length code. How many bits are required to encode the message FDEGAACAAGAAFABAAB? 2
(b) Build the Huffman coding tree for the message 'data structure'. 5.5
(c) What are the advantages and disadvantages of threaded binary tree? 2.5
4. (a) What is collision resolution? Name some collision resolution techniques. 2
(b) Draw a hash table with chaining and a size of 9. Use the hash function " $k \% 9$ " to insert the keys 5, 29, 20, 0, 27, 26, 17 and 18 into your table. 3
(c) The keys 12, 18, 13, 2, 3, 23, 5 and 15 are inserted into an initially empty hash table of length 10 using open addressing with hash function $h(k) = k \bmod 10$ and linear probing. What is the resultant hash table? Justify your answer. 5

0	
1	12
2	2
3	23
4	13
5	15
6	3
7	5
8	18
9	

(A)

0	3
1	2
2	12
3	13
4	23
5	5
6	15
7	
8	18
9	

(B)

0	
1	
2	12
3	13
4	2
5	3
6	23
7	5
8	18
9	15

(C)

0	
1	
2	12, 2
3	13, 3, 23
4	
5	5, 15
6	
7	
8	18
9	

(D)

SECTION B

5. (a) Differentiate between stack and queue. 2
 (b) Convert the following infix expression to its equivalent prefix and postfix expression 4
 i. $A + (B * C / 5 - (D / E ^ F) * G) * H$
 ii. $(1/2) * 3 + 4 ^ 5 + 6 * 7$
 (c) Simulate the postfix expression evaluation algorithm on the data 18, 6, /, 6, 2, +, *, 12, 4, /, - by showing stack's contents as each element is scanned. 4
6. (a) Define the terms (i) siblings (ii) successor and (iii) level. 3
 (b) Travel the tree in figure 6 in preorder, inorder and postorder. 3
 (c) How can you search '14' in the following binary search tree (figure 6)? However, if '5' is not in the tree just insert to its appropriate place and show the resultant tree. 4

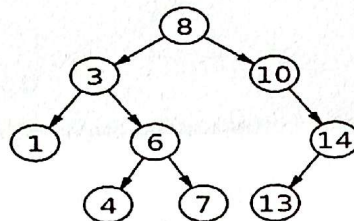
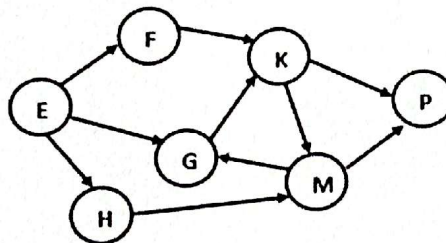


Figure: (6)

7. (a) Suppose the following six numbers are inserted in order into an empty binary search tree. 55, 77, 50, 83, 45, 20. Show the tree as each number is inserted into a binary search tree. 3
 (b) Discuss the linked representation of binary tree in memory. 3.5
 (c) Simulate the maxheap algorithm for the following values: 88, 56, 95, 84, 56, 48, 90, 19, and 12. 3.5
8. (a) How many ways a graph G can be traversed? What is the significance of the STATUS field? 2
 (b) What is adjacency matrix? How is it formed? 2
 (c) Consider the following graph that represents the daily flights between cities of some airline, and use DFS to find the path if someone want to fly from city E to city K with minimum no. of stops. 6



University of Rajshahi (Affiliated College)
Department of Computer Science and Engineering
B. Sc. (Engg.) Part-II Odd Semester Examination 2022
Course: CSE 2121 (Data Structure) (Syllabus: 2020-2021)
Full Marks: 60 Duration: 3 (Three) Hours

Answer 06 (Six) questions taking any 03 (Three) questions from each part

Section-A

1. a) Distinguish between linear and non-linear data structure. 2
b) Suppose you are given the following data: 10 30 50 78 90 95 100 105 110. Simulate the binary search algorithm for searching the item 110. 4
c) Maze (2:6, -4:-1, 1:5) is a 3D array with base=100, w=4. Calculate Maze [5, -1, 4] address in a row of major order and column major order. 4
2. a) What is vector in STL? 2
b) Write a program to create a graph using vector. 4
c) How can you find out connected components of a graph? 4
3. a) How can you locate element a_{ij} of a sparse matrix from a 1D array? 3
b) Convert the following infix expression to its equivalent prefix and postfix expression: 5
 i. $A + (B - C + D) * E / F + G \uparrow H$
 ii. $1 + 2 - (3 * 4 / 5 \uparrow 6) * 7$
c) Suppose one array contains 10 elements and another 10 elements are there in a Linked list. Which methods take longer time to access 5th element? Justify your answer. 2
4. a) Explain how a recursive function can be used to delete kth node from linked list. 3
b) Briefly discuss different hash collision recursion techniques. 3
c) Draw a hash table with chaining and a size of 9. Use a hash function "k%9" to insert the keys 5, 29, 20, 0, 27, 26, 17, and 18 into your table. 4

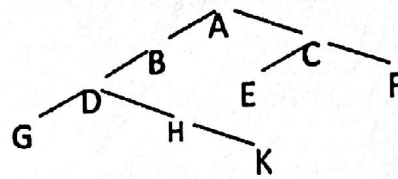
Section-B

5. a) What in binary indexed tree? 2
b) Why do you need to use binary indexed tree? 4
c) How can you find out parent of a node in binary indexed tree? 4
6. a) A complete binary tree has 1129 nodes, find out the depth of the tree. 3
b) Distinguish between maxheap and minheap. 3
c) Simulate the maxheap algorithm for the values: 67, 29, 90, 12, 34, 90, 9, and 12. 4
7. a) Suppose the following numbers are inserted inorder into an empty binary search tree: 40, 60, 50, 33, 35, 55, 11, and 12. Show the tree as each number is inserted into a binary 4

search tree.

- b) What will be the depth of a complete binary tree with 10,00,000 nodes.
c) Traverse the following tree in preorder, inorder, and postorder.

1.5
4.5



8. a) Use the Warshall's algorithm to find the shortest path matrix of the adjacency matrix 4

given below: $A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$.

- b) Consider the adjacency list of the Graph G in the following table. Find the nodes that are reachable from node C using Depth First Search. 4

Node	Adjacency	Node	Adjacency
A	G, E	E	C
B	C	F	A, B
C	F	G	B, C, E
D	C	H	D

- c) How many ways a graph G can be traversed? What is the significance of the STATUS field? 2

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Course: CSE 2121 (Data Structure) (Syllabus: 2019-2020)
Full Marks: 52.5 Duration: 3 (Three) Hours

6%

Answer 06 (Six) questions taking any 03 (Three) questions from each part

Section-A

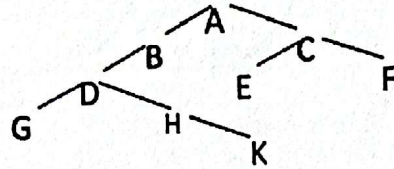
1. a) Write examples of linear and non-linear data structure. 1.75
b) For given data: 10 30 50 78 90 95 100 105 110, write the steps of the binary search algorithm for searching the item 90. 3.5
c) Maze (2:6, -4:-1, 1:5) is a 3D array with base=100, w=4. Calculate Maze [5, -1, 4] address in a row of major order and column major order. 3.5
2. a) Mention the advantages of Linked list over arrays. 2
b) State how an item can be inserted at the beginning, after a given node, at the end, and to a sorted Linked list. 4
c) Draw the diagrams of different types of Linked list. 2.75
3. a) How can you locate element a_{ij} of a sparse matrix from a 1D array? 2
b) Convert the following infix expression to its equivalent prefix and postfix expression: 4.75
 i. $A + (B - C + D) * E / F + G \uparrow H$
 ii. $1 + 2 - (3 * 4 / 5 \uparrow 6) * 7$
c) Suppose one array contains 10 elements and another 10 elements are there in a Linked list. Which methods take longer time to access 5th element? Justify your answer. 2
4. a) Explain how a recursive function can be used to delete kth node from linked list. 2.75
b) Briefly discuss different hash collision recursion techniques. 2
c) Draw a hash table with chaining and a size of 9. Use a hash function "k%9" to insert the keys 5, 29, 20, 0, 27, 26, 17, and 18 into your table. 4

Section-B

5. a) Discuss the array representation process of Stack with example? 2.75
b) Simulate the postfix expression algorithm using 18, 6, /, 6, 2, /, +, 3, -, 4, * by showing Stack's contents as each element is scanned. 4
c) What are the benefits of polish notation? 2
6. a) A complete binary tree has 1129 nodes, find out the depth of the tree. 2.25
b) Distinguish between maxheap and minheap. 2
c) Simulate the maxheap algorithm for the values: 67, 29, 90, 12, 34, 90, 9, and 12. 4.5

7. a) Suppose the following numbers are inserted inorder into an empty binary search tree: 40, 60, 50, 33, 35, 55, 11, and 12. Show the tree as each number is inserted into a binary search tree. 4.25

b) Traverse the following tree in preorder, inorder, and postorder. 4.5



8. a) What are the differences between complete binary tree and extended binary tree? 2
- b) How can you represent a tree in memory using a linear array? 3
- c) Suppose the preorder traversal of a binary tree T yields: $* + a - bc / - de - + fgh$. Draw the tree T and perform inorder and postorder traversal. 3.75

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Course: CSE-2121 (Data Structure)

Full Marks: 52.5

Duration: 3Hours

[Answer six questions taking any three from each section]

Section A

1. (a) What is the benefit of binary search over linear search? 2
(b) Maze (1:6, -4:1, 5:10) is a 3D array with base=100, w=4, calculate Maze [3, -2, 8] address 3.75
in a row major order and column major order. 3
(c) We often store sparse matrix in a 1D array to save spaces. What is the memory saving if
we store a sparse matrix in a 1D array rather than a 2D array? 3
2. (a) What is a linked list? Explain the main differences between the linked list and linear array? 2.75
(b) Briefly discuss the terms garbage collection, overflow and underflow. 3
(c) One of the advantages of linked list is the ability to insert data into the list easily. Explain 3
with your own words and figures how to insert data at the beginning, after a given node,
at the end and to a sorted list.
3. (a) Define stack. Explain the usage of stack in recursive algorithm implementation. 3
(b) Simulate the infix to postfix transformation algorithm for 3.25
Q: $A + (B * C - (D / E \uparrow F) * G) * H$ by showing the stack's contents as each element is
scanned. 2.5
(c) Write down a routine to insert an element onto a queue.
4. (a) What is a hash function? 2
(b) The keys 14, 12, 18, 13, 2, 3, 23, 5 and 15 are inserted into an initially empty hash table 3
of length 10 using open addressing with hash function $h(k) = k \bmod 10$ and plus 3 probing.
What is the resultant hash table? 3.75
(c) A hash table of length 10 uses open addressing with hash function $h(k) = k \bmod 10$, and
linear probing. After inserting 6 values into an empty hash table, the table is as shown
below.

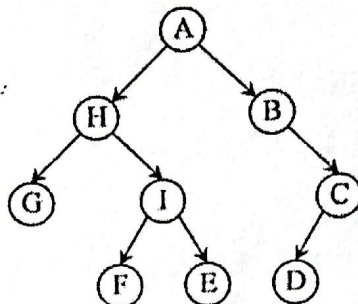
		42	23	34	52	46	33		
0	1	2	3	4	5	6	7	8	9

Which one of the following choices gives a possible order in which the key values could have been inserted in the table?

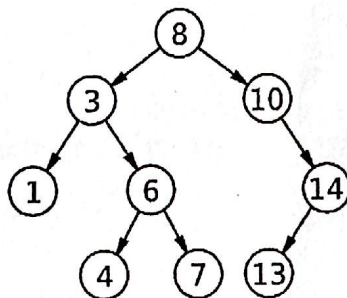
- (A) 46, 42, 34, 52, 23, 33
(B) 34, 42, 23, 52, 33, 46
(C) 46, 34, 42, 23, 52, 33
(D) 42, 46, 33, 23, 34, 52

Section B

5. (a) Define the terms (i) siblings, (ii) ancestor, and (iii) depth of a binary tree. 3
 (b) Tree traversal (also known as walking the tree) refers to the process of visiting each node exactly once. Simulate the preorder traversal algorithm for the following tree. 5.75



6. (a) What is adjacency matrix? How is it formed? 2.75
 (b) Define the terms (i) Isolated node, (ii) Simple Path and (iii) Weighted Graph. 3
 (c) How can you search '5' in the following binary search tree? However if '5' is not in the tree just insert to its appropriate place and show the resultant tree. 3



7. (a) What is meant strongly and weakly connected in a graph? 2
 (b) Prove that the maximum number of edges that a graph with n vertices is $n*(n-1)/2$. 2.75
 (c) Explain Breadth First search algorithm with example. 4
8. (a) Suppose we want to encode a message constructed from the symbols A, B, C, D, E, F and G using a fixed-length code. How many bits are required to encode the message FDEGAACAAGAAFABA? 2
 (b) Build the Huffman coding tree for the message 'science engineering'. 3.75
 (c) Suppose you are to insert a node 'X' as the right child of node 'P'. Discuss the inserting mechanisms with figures. 3
- i. When the right subtree of 'P' is empty.
 - ii. When the right subtree of 'P' is not empty.

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Time: 3 Hours

Full Marks: 52.5

[Answer six questions taking any three from each section]

Section A

1. a) How can you differ linear data structures from non-linear data structures? 2.00
b) Simulate the binary search algorithm on the following data: 10 30 50 78 90 95 100 4.00
105 110 (suppose we search for item 110). 2.75
c) How are the elements of a 2D array stored in the memory?
2. a) *Score* is a 20x5 matrix with base address 200 and $w=4$. Find out the address of the 4.00
element *Score*[10,2] for both row major order and column major order.
b) What is sparse matrix? What is the difference between triangular matrix and 3.00
tridiagonal matrix?
c) How can you locate element a_{ij} of a sparse matrix from a 1D array? 1.75
3. a) Mention some differences between arrays and linked lists. 1.75
b) Can binary search algorithm be employed on data stored in linked list? Justify 2.00
your answer.
c) Briefly discuss inserting mechanism of an item at the beginning, after a given 5.00
node, at the end and also to a sorted list.
4. a) What is hash table? What can be the techniques to avoid collision? 2.75
b) Draw a hash table with chaining and a size of 9. Use the hash function " $k\%9$ " to 3.00
insert the keys 5, 29, 20, 0, 27, 26, 17 and 18 into your table.
c) A hash table of length 10 uses open addressing with hash function $h(k)=k \bmod 10$, 3.00
and linear probing. After inserting 6 values into an empty hash table, the table is as
shown below:

0	1	2	3	4	5	6	7	8	9
		42	23	34	52	46	33		

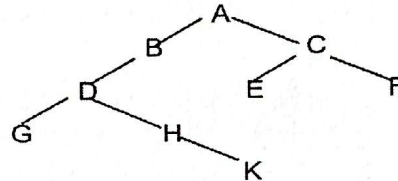
Which one of the following choices gives a possible order in which the key values could have been inserted in the table? Explain.

- i. 46, 42, 34, 52, 23, 33
- ii. 34, 42, 23, 52, 33, 46
- iii. 46, 34, 42, 23, 52, 33
- iv. 42, 46, 33, 23, 34, 52

Section B

5. a) What is a queue, how is it different from a stack and how is it implemented? 3.00
 b) Write the steps involved in the insertion and deletion of an element in a stack. 3.00
 c) Simulate the postfix expression evaluation algorithm using **18, 6, /, 6, 2, /, +, 3, -, 4, *** by showing Stack's contents as each element is scanned. 2.75

6. a) When do we call a binary tree to be a 2-tree? What is the depth of a complete binary tree with 10,00,000 nodes? 2.00
 b) Traverse the following tree in preorder, inorder and postorder. 3.00



- c) How can binary search tree overcome the disadvantages of array and linked list? Explain the process of inserting a new unique item in a binary search tree with a simple example. 3.75
7. a) Suppose the following eight numbers are inserted in order into an empty binary search tree: 40, 60, 50, 33, 35, 55, 11, 12. Show the tree as each number is inserted into a binary search tree. 4.00
 b) Consider the following adjacency matrix below: 4.75

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

Now find out A^2 , A^3 , A^4 , B_4 and from that make the path matrix and tell whether this is strongly connected or not.

- 8) a) What is directed graph? 1.00
 b) Discuss the sequential representation of graph with example. 4.00
 c) Use the Warshall's algorithm to find the shortest path matrix of the adjacency matrix given below: 3.75

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

[Answer six questions taking any three from each section]

Section A

- | | | |
|----|--|------|
| 1. | (a) Define data structure. Why data structure is necessary? | 3 |
| | (b) What do you mean by Linear data structure and Nonlinear data structure? Give example. | 2.75 |
| | (c) What is 2D array? How can you represent 2D array in memory? | 3 |
| 2. | (a) Suppose Book(2:10, -4:-1, 5:8) is a 3D array with base=200, w=4, calculate Book[5, -1, 7] address in a row major order and column major order. | 3.75 |
| | (b) What is pointer array? How a pointer can save memory space when storing a 2D array? | 4 |
| | (c) Is there any differences between a record and a linear array? | 1 |
| 3. | (a) What are the advantages of Linked List over array? | 2.75 |
| | (b) Briefly discuss inserting mechanism of an item at the beginning, after a given node and at the end. | 3 |
| | (c) What is two way lists? Why it is important? Explain with Schematic diagram. | 3 |
| 4. | (a) Briefly discuss different hash collision resolution techniques. | 2.75 |
| | (b) Draw a hash table with open addressing and a size of 9. Use the hash function "k%9" and quadratic probing. Insert the keys: 5, 29, 20, 0, 27, 26, 17 and 18 into your table. | 3 |
| | (c) Given the following input (4322, 1334, 1471, 9679, 1989, 6171, 6173, 4199) and the hash function $x \text{ mod } 10$, which of the following statements are true? Explain. | 3 |
| | i. 9679, 1989, 4199 hashes to the same value | |
| | ii. 1471, 6171 has to the same value | |
| | iii. All elements hash to the same value | |
| | iv. Each element hashes to a different value | |
| | v. None of the above. | |

Section B

- | | | |
|----|--|------|
| 5. | (a) Differentiate between stack and queue. | 1.75 |
| | (b) Convert the following infix expression to its equivalent prefix and postfix expression.
$A + (B * C / 5 - (D / E \uparrow F) * G) * H$ | 3 |
| | (c) Simulate the postfix expression evaluation algorithm using 18, 6, /, 6, 2, +, *, 12, 4, /, - by showing Stack's contents as each element is scanned. | 3 |
| | (d) What is deque? | 1 |
| 6. | (a) What is complete binary tree? What is the parent child relationship? | 2 |
| | (b) Discuss the linked representation of binary tree in memory. | 3 |
| | (c) Simulate the maxheap algorithm for the following values: 100, 29, 90, 148, 12, 34, 90, 89, 120, 99 and 12. | 3.75 |
| 7. | (a) What is Directed Graph? Explain. | 2 |
| | (b) Briefly discuss the advantages of binary search tree over array and linked list. | 2 |
| | (c) Use the Warshall's algorithm to find the shortest path matrix of the weighted matrix given below. | 4.75 |
| | $W = \begin{pmatrix} 2 & 5 & 0 & 0 \\ 2 & 0 & 1 & 4 \\ 5 & 3 & 3 & 3 \\ 4 & 2 & 3 & 0 \end{pmatrix}$ | |
| 8. | (a) Discuss the sequential Representation of Graph with example. | 3 |
| | (b) How many ways a graph G can be traversed? What is the significance of the STATUS field? | 2.75 |
| | (c) Consider the adjacency list of the Graph G in the following table. Find the nodes that are reachable from node C using Depth First Search. | 3 |

Node	Adjacency	Node	Adjacency
A	G, E	E	C
B	C	F	A, B
C	F	G	B, C, E
D	C	H	D

University of Rajshahi
Department of Computer Science and Engineering
B.Sc. (Engg.) Part-2 (Odd Semester) Examination-2018
Course: CSE2121 (Data Structure)
Marks: 52.50 Time: 3:00 Hours

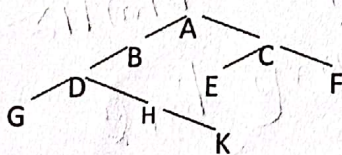
[N.B. Answer any Six questions taking Three from each section.]

Section-A

1. a) How can you sort the elements of a matrix? Explain. 1.75
b) How does a pointer array can save memory when store a variable sized group of data? Discuss 4
with necessary figures.
c) Maze(2:6, -4:-1, 1:5) is a 3D array with base=100, w=4. Calculate Maze[5, -1, 4] address in a 3
row major order and column major order.
2. a) What is linked list? Discuss different types of linked list with diagram. 3
b) Discuss the advantages and disadvantages of linked list? 4
c) What is Garbage collection? When does it take place? 1.75
3. a) What is sparse matrix? How can you locate element a_{ij} of a sparse matrix from a 1D array? 3
b) Discuss the array representation mechanism of stack. 1.75
c) Convert the following infix expression to its equivalent prefix and postfix expression. 4
i) $A+(B-C+D)*E/F+G\uparrow H$
ii) $1+2-(3*4/5\uparrow 6)*7$
4. a) What is polish notation? What are the benefits of polish notation? 1.75
b) Simulate the infix to postfix transformation algorithm for $Q: A + (B * C - (D / E \uparrow F) * G) * H$ 4
by showing the stack's contents as each element is scanned.
c) Suppose 15 elements are maintained by array and another 15 are by Linked List. Which 3
methods take longer time to access 9th element. Justify your answer.

Section-B

5. a) Define the following terms: siblings, successor, and level. 3
b) What is binary search tree. How can binary search tree overcome the disadvantage of array and 2.75
linked list?
c) Traverse the following tree in preorder, inorder and postorder. 3



6. a) Discuss the linked representation of Graph with example. 2.75
b) Define the following terms with example: 3
(i) Strongly Connected Graph (ii) Directed Graph (iii) Weighted Graph
c) Consider the following adjacency matrix below: 3

$$A = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

Now find out A^2 , A^3 , A^4 , B^4 and from that make the path matrix and tell whether this is strongly connected or not.

7. a) What is heap? What are the advantages of heap?
 b) Use the Warshall's algorithm to find the shortest path matrix of the weighted matrix given below.

$$W = \begin{pmatrix} 2 & 4 & 1 & 0 \\ 3 & 0 & 0 & 5 \\ 5 & 1 & 3 & 6 \\ 3 & 2 & 3 & 0 \end{pmatrix}$$

- c) Suppose the following six numbers are inserted in order into an empty binary search tree. 10, 30, 50, 49, 33, 55, 13. Show the tree as each number is inserted into a binary search tree. 3
8. a) Define hash function and hash table and load factor. 2.75
 b) Briefly discuss open addressing quadratic probing method with example.
 c) Draw a hash table with chaining and a size of 9. Use the hash function " $k \% 9$ " to insert the keys 5, 29, 20, 0, 27, 26, 17 and 18 into your table. 3

University of Rajshahi
Department of Computer Science and Engineering
B. Sc. (Engg.) Part-2 Odd Semester Examination-2017
Course: CSE-2121 (Data Structure)
Full Marks: 52.5 Time: 3:00 Hours

[N.B. Answer any SIX questions taking THREE questions from each part]

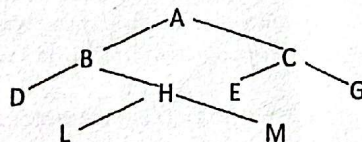
Part-A

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1. a) What is linear array? How can you find the no. of elements in any linear array? 3.75
b) Simulate the binary search algorithm on the following data: 11 22 33 44 55 66 77 88 99 110 121 132 143 (suppose we search for item 120).
c) What is 2D array? How can you represent 2D array in memory? 3
2. a) Room (1:8, -4:1, 6:10) is a 3D array with base=400, w=2, calculate Room [4, -2, 7] address in a row major order and column major order. 3
b) What is sparse matrix? How can you locate element a_{ij} of a sparse matrix from a 1D array? 2.75
c) What is memory saving if we store a sparse matrix in a 1D array rather than a 2D array? 3
3. a) What is a Linked List? Discuss with example. 1
b) Define overflow and underflow? How can you handle them? 1.75
c) Briefly discuss inserting mechanism of an item at the beginning, after a given node, at the end and to a sorted list. 5
d) What is Header Linked List? 1
4. a) Differentiate between stack and queue. 1.75
b) Convert the following infix expression to its equivalent prefix and postfix expression . 4
(i) $A+(B-C)*D/E+F/G \uparrow H$
(ii) $(1+2)*3-4/5*6 \uparrow 7$
c) Simulate the postfix expression evaluation algorithm using 20, 5, 2, *, /, 2, 3, ↑, 4, /, - by showing Stack's contents as each element is scanned. 3

Part-B

5. a) What is complete binary tree? A complete binary tree has 129 nodes, find out the depth of the tree. 2
b) Discuss the linked representation of binary tree in memory. 2.75
c) Simulate the maxheap algorithm for the following values: 67, 29, 90, 48, 12, 34, 90, 9, and 12. 4
6. a) Illustrate similar and copies with example. 2
b) What is an extended tree? 1
c) Simulate the postorder traversal algorithm for the following tree. 5.75



7. a) Define the following terms: Connected Graph, Path, and Weighted Graph. 3
b) How many ways a graph G can be traversed? What is the significance of the STATUS field? 2
c) Consider the adjacency list of the Graph G in the following table. Draw the graph and find out the path from A to F with minimum number of nodes along that path using Breadth First Search. 3.75

Node	Adjacency	Node	Adjacency
A	E, G	E	H
B	C	F	A, B
C	F	G	B, C, E
D	C	H	D

8. a) Define deque and priority queue with example. 2
b) Write the steps of preorder and postorder traversal of a binary tree. 4
c) What is binary search tree? Mention the advantages of a binary search tree. 2.75

Answer 06(Six) questions taking any 03(Three) questions from each part

Part-A

1. (a) Define data structure. Why is data structure necessary? 3
 (b) What do you mean by Linear data structure and Nonlinear data structure? Give example. 2.75
 (c) What is sparse matrix? What is the difference between triangular matrix and Tridiagonal matrix? 3

2. (a) What are the advantages of Linked List? 4
 (b) Suppose 10 elements are maintained by array and another 10 are by Linked List. Which methods take longer time to access 7th element. Justify your answer. 2
 (c) What is two way lists? Why is it important? Explain with schematic diagram. 2.75

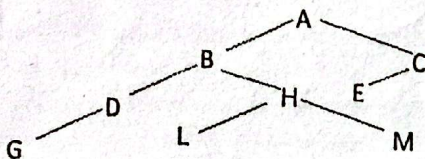
3. (a) Discuss the array representation mechanism of stack. 3
 (b) What is polish notation? What are the benefits of polish notation? 2
 (c) Convert the following infix expression to its equivalent prefix and postfix expression. 3.75
 (i) $A*B/C+D\uparrow(E-F*G)/H$
 (ii) $1+2*3/4\uparrow5*6-7*8$

4. (a) Simulate the postfix expression evaluation algorithm using 12, 6, /, 6, 2, +, *, 12, 4, /, - by showing Stack's contents as each element is scanned. 3
 (b) What is recursion? Explain the use of recursion. 3
 (c) Explain the operations on queue with example. 2.75

Part-B

5. (a) Briefly discuss inserting and deleting mechanism of an item in the linked list. 4.75
 (b) Define deque and priority queue with example. 2
 (c) What is garbage collection? When does it take place? 2

6. (a) Simulate the preorder traversal algorithm for the following tree. 5



- (b) What is binary search tree? Why binary search tree is important? 2
 (c) What is the difference between maxheap and minheap? 1.75

7. (a) Define weighted graph and directed graph with example.
 (b) Consider the following adjacency matrix below:

2.75

3

$$A = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

Now find out A^2 , A^3 , A^4 , B^4 and from that make the path matrix and tell whether this is strongly connected or not.

- (c) Use the Warshall's algorithm to find the shortest path matrix of the weighted matrix given below. 3

$$W = \begin{pmatrix} 6 & 8 & 0 & 0 \\ 3 & 0 & 0 & 9 \\ 5 & 8 & 3 & 6 \\ 6 & 2 & 3 & 0 \end{pmatrix}$$

8. (a) Discuss the sequential representation of Graph with example.
 (b) Consider the adjacency list of the Graph G in the following table. Find the nodes that are reachable from node C using Depth First Search.

4

4.75

Node	Adjacency	Node	Adjacency
A	G, E	E	C
B	C	F	A, B
C	F	G	B, C, E
D	C	H	D

Part -A

1. a) Why data structure is needed? Differentiate between linear and nonlinear data structure. 3
b) Explain linear array representation in memory. 2
c) Simulate the binary search algorithm on the following data: 10, 20, 50, 70, 80, 90, 100, 110 (suppose we search for item 110). 3.75
2. a) For column major order find out the address of the element score [15,3] from a 20X5 matrix array score with base value 100 and w=4. 3
b) How does a pointer array can save memory when stores a variable sized group of data? Discuss with necessary figures. 3.75
c) What is a record? What is the difference between a record and a linear array? 2
3. (a) Explain the representation of linked lists in memory. 3
(b) Discuss header linked list. Describe grounded and circular header list. 3
(c) Briefly explain two-way linked list. 2.75
4. a) What is stack? What are the operations on stack? Explain with example. 3
b) Convert the following infix expression to its equivalent prefix and postfix expression 3.75
(i). $A+B*C/D-E+(F/G+H \uparrow K)$
(ii). $(1+2) \uparrow 3/4*5+7-8 \uparrow 9$
c) What is priority queue? Why is it important? 2

Part-B

5. a) Illustrate similar and copies of a tree with examples. 3
b) What is complete binary tree? What is the parent-child relationship? 2
c) For the expression: $*+a-bc*-de-/fgh$ draw the tree and perform inorder and postorder traversal. 3.75
6. (a) What is binary search tree? 1.75
(b) Suppose the following six numbers are inserted into an empty binary search tree: 33, 50, 45, 52, 12, 10. Show the tree as each number is inserted into a binary search tree. 4
(c) Simulate the maxheap algorithm for the following values: 77, 40, 90, 65, 20, 35, 95, 10, 15. 3
7. a) Define the following terms: (i) Degree of a node, (ii) Isolated node, (iii) Path, (iv) Multi Graph. 3.75
b) Consider the following adjacency matrix below: 5

$$A = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix}$$

Now find out A^2 , A^3 , A^4 , B^4 and from that make the path matrix and tell whether this graph is strongly connected or not.

8. a) Discuss the linked representation of Graph with example. 3.75
b) Consider the adjacency list of the Graph G in the following table. Draw the graph and find out the path from A to H with minimum number of nodes along that path using Breadth First Search. 5

Node	Adjacency	Node	Adjacency
A	E, G	E	H
B	C	F	A, B
C	F	G	B, C, E
D	C	H	D

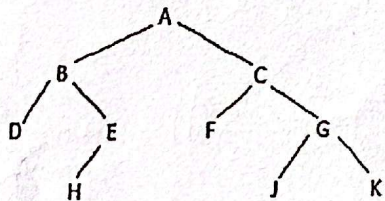
Answer six (06) questions taking three (03) from each part

Part A

1. a) What is data structure? What are the differences between linear and nonlinear data structure? 2.5
b) Explain the depth-first search technique. 3.5
c) What do you mean by internal and external sorting? List some names of each type. 2.75
2. a) What is a linked List? Discuss with example. 2
b) Suppose 20 elements are maintained by array and another 10 are by Linked List. Which method will take longer time to access 9th element? Justify your answer. 2
c) Briefly discuss inserting mechanism of an item at the beginning, after a given node and at the end. 4.75
3. a) What is polish notation? What are the benefits of polish notation? 2
b) Convert the following infix expression to its equivalent prefix and postfix expression: 4
(i) $A * B + (C * D / E) * F + (G \uparrow H)$
(ii) $1 * 2 / 3 + (4 - 5 \uparrow 6) + 7 - 8$
c) Simulate the postfix expression evaluation algorithm using 12, 6, /, 6, 2, +, *, 12, 4, /, - by showing Stack's contents as each element is scanned. 2.75
4. a) What is overflow and underflow? How can you handle them? 3.25
b) What is a two way list? Why is it important? Explain with schematic diagram. 3.5
c) What is garbage collection and compaction? 2

Part B

5. a) For column major order find out the address of the element marks[12,3] from a 25X4 matrix array marks with base value 250 and w=8. 2
b) What is pointer and pointer array? 1.75
c) How a pointer can save memory space to store a 2D array? 3
d) What is the difference between Triangular matrix and Tridiagonal matrix? 2
6. a) Define heap, leaf and depth of tree. For 5009 nodes, find out the depth of the tree. 3.75
b) Discuss the linked representation of binary tree in memory. 2
c) Simulate the preorder traversal algorithm for the following tree 3



7. a) Define the following graph terms: (i) Adjacent Nodes, (ii) Cycle, (iii) Connected graph, and (iv) Weighted graph. 2
b) Discuss the sequential Representation of Graph with example. 2.75
c) Consider the following adjacency matrix below: 4

$$A = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \end{pmatrix}$$

Now find out A^2 , A^3 , A^4 , B_4 and from that make the path matrix and tell whether this is strongly connected or not.